Requirements Specification

ShopSense: the personalised shopping research agent

Document	Requirements Specification
Project	ShopSense: the personalised shopping research agent
Course	IT314 Software Engineering
Team	Group 13 (10 members)
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This document lists all 55 requirements of ShopSense: 28 functional, 17 non-functional and 10 domain requirements. For every requirement it states its priority, the sprint or sprints in which it is built, and the user stories that build it.

1. How to read this document

- **Priority** follows the MoSCoW method. **Must** means the product cannot work without it. **Should** means it is important, but the product still works without it. A requirement takes the highest priority of the user stories that build it.
- **Sprint(s)** gives the sprint or sprints in which the requirement is built. There are six sprints of two weeks each, from Sprint 1 (21 September 2026) to Sprint 6 (13 December 2026).
- **User stories** gives the stories from the product backlog that build the requirement. They are written out in full in the User Stories document.
- Every requirement is built by at least one user story, and every user story traces back to at least one requirement.

Summary

Requirement type	Count	Must	Should
Functional (FR)	28	20	8
Non-functional (NFR)	17	14	3
Domain (DR)	10	10	0
Total	**55**	**44**	**11**

2. Functional requirements

Functional requirements describe what the system must do.

ID	Requirement	Priority	Sprint(s)	User stories
FR-01	The system shall allow a user to enter a shopping request using natural language.	Must	1	US-01
FR-02	The system shall identify the product category, intended use, budget, and explicit constraints from the user's request when present.	Must	1	US-02

ID	Requirement	Priority	Sprint(s)	User stories
FR-03	The system shall determine the evaluation criteria relevant to the detected product category.	Must	1	US-03
FR-04	The system shall generate an adaptive set of interview questions based on the category and information already known about the user.	Should	4	US-04
FR-05	The system shall ask interview questions sequentially and adapt later questions according to previous answers.	Must	1	US-05
FR-06	The system shall reuse relevant information from previous research sessions to avoid unnecessary or repetitive questions.	Should	5	US-06
FR-07	The system shall explain unfamiliar technical terms when the user requests clarification during the interview.	Should	6	US-07
FR-08	The system shall interpret qualitative responses such as “low,” “medium,” “high,” or “good enough” into appropriate preference values or categories.	Must	2	US-08
FR-09	The system shall distinguish mandatory requirements from optional preferences and construct a personalized evaluation rubric.	Must	3	US-09
FR-10	The system shall allow users to specify unwanted brands, products, features, or other exclusion criteria.	Must	3	US-10
FR-11	The system shall generate research queries and retrieve relevant information from configured sources such as Reddit, expert reviews, YouTube, search services, and product information sources.	Must	2, 5	US-12, US-38
FR-12	The system shall extract relevant product mentions, claims, opinions, and supporting evidence from retrieved research material.	Must	2	US-14
FR-13	The system shall normalize product names, aliases, model numbers, and variants into identifiable product entities.	Must	3	US-15
FR-14	The system shall verify important product specifications and identify conflicting or insufficient information across relevant sources.	Must	3	US-16
FR-15	The system shall filter candidate products that violate mandatory user constraints before personalized ranking.	Must	3	US-17
FR-16	The system shall calculate and rank personalized scores for eligible products and provide a confidence value reflecting evidence quality and coverage.	Must	3, 4	US-18, US-19
FR-17	The system shall provide an explanation showing how user priorities, product characteristics, and evidence contributed to the recommendation.	Must	4	US-20
FR-18	The system shall provide citations and source references for important factual claims and recommendation evidence.	Must	3	US-21
FR-19	The system shall allow users to compare shortlisted products and dynamically rerank them when preference weights or constraints are changed.	Must	5, 6	US-23, US-37
FR-20	The system shall retrieve available price, availability, and purchase-link information and associate it with the corresponding recommended product or variant.	Must	5	US-24
FR-21	The system shall check each sentence of a generated explanation against the evidence it cites, and remove or flag unsupported sentences.	Should	5	US-22

ID	Requirement	Priority	Sprint(s)	User stories
FR-22	The system shall let a user view, edit, export and delete their stored preferences.	Should	6	US-11
FR-23	The system shall evaluate the quality of Reddit threads and comments before using them as evidence.	Should	2, 6	US-13, US-36
FR-24	The system shall look up missing high-importance facts for the top-ranked products before estimating them.	Must	5	US-41
FR-25	The system shall recommend only products supported by at least one retrieved source.	Must	3	US-40
FR-26	The system shall recognise interview replies that are questions, skips, corrections or unclear, and respond to each appropriately.	Must	4	US-39
FR-27	The system shall account for negation, sarcasm, complaint strength and disagreement between communities when reading community posts.	Should	5	US-42
FR-28	The system shall reuse a recent result when a new search has the same request, category and priorities.	Should	6	US-34

3. Non-functional requirements

Non-functional requirements describe how well the system must work: its speed, reliability, security, usability and quality.

ID	Category	Requirement	Priority	Sprint(s)	User stories
NFR-01	Performance	The system shall provide responses to normal user interactions within an acceptable response time under expected operating conditions.	Should	6	US-34
NFR-02	Performance	Independent research operations should be capable of executing concurrently where appropriate to reduce overall research time.	Must	2	US-27
NFR-03	Reliability	The system shall gracefully handle external API failures, timeouts, invalid responses, and temporary service unavailability.	Must	4, 6	US-31, US-50, US-53, US-54
NFR-04	Reliability	The system shall support fallback to an alternative configured LLM or service when the primary service is unavailable.	Must	1, 2	US-26, US-46
NFR-05	Reliability	The system shall validate structured responses received from external LLMs before using them in subsequent processing.	Must	1, 2	US-26, US-48
NFR-06	Security	The system shall authenticate users before providing access to protected preferences and research history.	Must	4	US-30
NFR-07	Security	The system shall prevent unauthorized users from accessing another user's personal preferences or research sessions.	Must	4	US-30
NFR-08	Security	API keys, authentication credentials, and other sensitive configuration values shall not be exposed in application source code.	Must	1	US-25

ID	Category	Requirement	Priority	Sprint(s)	User stories
NFR-09	Security	Content retrieved from external sources shall be treated as untrusted data and shall not be allowed to override system instructions or security controls.	Must	2	US-28
NFR-10	Usability	The system shall present interview questions, explanations, and recommendation results in language understandable to non-technical users.	Must	6	US-33
NFR-11	Accessibility	The interface shall provide accessible controls for interview questions, preference sliders, recommendation cards, and product comparisons.	Must	6	US-33
NFR-12	Maintainability	The system shall use modular components with clearly defined interfaces so that agents, retrieval methods, and external integrations can be modified independently.	Must	1, 6	US-35, US-52
NFR-13	Extensibility	The architecture shall allow additional product categories, research sources, and service providers to be added without major modification to the core recommendation engine.	Should	6	US-35
NFR-14	Traceability	Important factual claims and recommendation decisions shall be traceable to their supporting evidence and sources.	Must	6	US-29
NFR-15	Testability	Core components such as preference processing, constraint filtering, scoring, and ranking shall be independently testable without requiring live external LLM or research API calls.	Must	2	US-32
NFR-16	Security	The system shall limit request rates per user and per network address.	Should	6	US-47
NFR-17	Quality	Every change shall pass automated quality checks, including tests on recorded real outputs and evaluation thresholds, before it is merged.	Must	4, 6	US-49, US-51

4. Domain requirements

Domain requirements come from the shopping and product-research domain itself. They hold true for any product category.

ID	Requirement	Priority	Sprint(s)	User stories
DR-01	The system shall consider the user's stated budget when filtering and ranking products.	Must	3, 5	US-17, US-24
DR-02	Different product variants shall be distinguished when their price, specifications, configuration, or other relevant characteristics differ.	Must	3	US-15
DR-03	Official product specifications should be verified using manufacturer or other authoritative technical sources whenever possible.	Must	3	US-16
DR-04	Product price and availability information shall be associated with its retrieval time because such information can change over time.	Must	5	US-24

ID	Requirement	Priority	Sprint(s)	User stories
DR-05	Information obtained from community discussions shall be treated as user experience or opinion unless independently verified as a factual claim.	Must	2	US-14
DR-06	Different names, abbreviations, and model identifiers referring to the same product shall be resolved into a common product entity where appropriate.	Must	3	US-15
DR-07	Product evaluation criteria shall vary according to the product category and the user's intended use.	Must	1	US-03
DR-08	Mandatory product requirements shall be treated as hard constraints, while optional preferences shall influence the relative ranking of eligible products.	Must	3	US-09
DR-09	Missing product information shall not automatically be interpreted as negative evidence against a product.	Must	3	US-16, US-18
DR-10	Product recommendations shall consider the individual user's intended use, priorities, and constraints rather than relying solely on general popularity or generic best-product lists.	Must	3, 6	US-18, US-51

5. Requirements by sprint

This table shows which requirements each sprint works on. A requirement built over more than one sprint appears under each of those sprints.

Sprint	Planned dates	Requirements worked on	Count
Sprint 1	21 Sep – 4 Oct 2026	FR-01, FR-02, FR-03, FR-05, NFR-04, NFR-05, NFR-08, NFR-12, DR-07	9
Sprint 2	5 – 18 Oct 2026	FR-08, FR-11, FR-12, FR-23, NFR-02, NFR-04, NFR-05, NFR-09, NFR-15, DR-05	10
Sprint 3	19 Oct – 1 Nov 2026	FR-09, FR-10, FR-13, FR-14, FR-15, FR-16, FR-18, FR-25, DR-01, DR-02, DR-03, DR-06, DR-08, DR-09, DR-10	15
Sprint 4	2 – 15 Nov 2026	FR-04, FR-16, FR-17, FR-26, NFR-03, NFR-06, NFR-07, NFR-17	8
Sprint 5	16 – 29 Nov 2026	FR-06, FR-11, FR-19, FR-20, FR-21, FR-24, FR-27, DR-01, DR-04	9
Sprint 6	30 Nov – 13 Dec 2026	FR-07, FR-19, FR-22, FR-23, FR-28, NFR-01, NFR-03, NFR-10, NFR-11, NFR-12, NFR-13, NFR-14, NFR-16, NFR-17, DR-10	15